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Practical 0

* **Aim:-** Study & installation of python.

* **Theory:-** Python is a high-level interpreted, object-oriented, and open-source programming language developed by Guido Van Rossum and first released in 1991. It is known for its simple syntax, readability, and ease of use making it one of the most popular programming languages for beginners as well as professionals.

Python is extensively used in various fields such as Machine Learning, AI, Data Science, Web development, Automation, Scientific Computing, and Software Development. In Machine Learning, Python provides a rich ecosystem of libraries and frameworks that simplify data preprocessing, compatibility make it the preferred choice for ML applications.

Python offers several built-in data structures such as lists, tuples, dictionaries, and sets, along with support for object-oriented and functional programming. It also includes a package manager called pip, which allows users to install and manage external libraries required for ML projects.

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Some important python libraries in ML.

(i) NumPy :- performs mathematical computations and multidimensional arrays.
(ii) Pandas :- Used for data cleaning, manipulation and analysis.
(iii) Matplotlib :- Creates graphs and data visuals.
(iv) Scikit-learn :- Provides ML algorithms for classification, regression, clustering & model evaluation.
(v) Tensorflow & Pytorch :- Used for developing deep learning and neural network models.

* **Installation :-**

(1) Visit official python website and download the latest version of python.
(2) Run the installer and click install now and wait for installation to complete.
(3) Verify the installation by opening CMD & typing python --version.
(4) Python can be programmed using IDEs such as VS code, Pycharm or Jupyter.

Python provides a flexible environment for developing, testing and deploying ML models efficiently.

* **Conclusion:-** Python was successfully installed. Its simple syntax, powerful libraries & easy-to-use development environment make it an ideal programming language for ML.

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Practical 1

* **Aim:-** Study of Python Basic Libraries such as Statistics, Math, NumPy and Scipy.

* **Theory:-** Python is widely used in Machine learning, Data Science, Artificial Intelligence, and Scientific Computing because it provides powerful libraries that simplify mathematical and statistical operations. Instead of writing complex code, developers can use libraries such as Statistics, Math, NumPy and SciPy to perform calculations quickly and accurately.

(1) The Statistics library is a built-in Python module used for statistical analysis of numerical data. It provides functions such as mean, median, mode, variance, and standard deviation, which help summarize data and identify patterns. These functions are used in data analysis and ML.

(2) The Math library is another built-in module that provider mathematical functions like square root, logarithms, factorials, trigonometric functions, exponentiation and constants such as π and e. It is commonly used for solving mathematical problem and performing accurate calculations.

(3) NumPy is a powerful library for numerical computing. It introduces the N-dimensional array, which is faster and more efficient than Python lists. NumPy supports array manipulation, matrix operations, linear algebra, random number generation, and mathematical functions, making it one of the most important libraries for ML and Data Science.

(4) SciPy is a built on top of NumPy and provides advanced scientific computing features. It includes modules for optimization, integration, interpolation, statistics, signal processing, image processing, and linear algebra. SciPy is widely used in engineering, research and data analysis for solving complex scientific problems.

Together these libraries provides a strong foundation for mathematical, statistical and scientific computing. They improve the speed, accuracy and efficiency of Python programs and are essential tools for ML and DS applications.

* **Conclusion:-** The practical helped in understanding the basic Python libraries Statistics, Math, NumPy and SciPy and their application. These libraries make mathematical calculations, statistical analysis and scientific computing faster and more efficient, forming an essential foundation for ML and DS.

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Practical 2

Aim:- Write a program to implement Simple Linear Regression.

Theory:- Simple Linear Regression is a supervised machine learning algorithm that is used to study the relationship between two numerical variables. It is method that uses one independent variable to predict the value of dependent variable.

The independent variable is represented by X, while the dependent or predicted variable is presented by y. Simple linear regression finds the best-fit straight line between the given data points.

The equation of a simple linear regression line is :-

$$Y = b_0 + b_1 X$$

where :-

* Y = Predicted value of the dependent var.
* X = Value of the Independent variable
* b_0 = Intercept of the regression line
* b_1 = Slope or regression coefficient.

The slope represents how much the dependent variable changes when the independent variable is increased by one unit.

The regression line is selected in such a way that the difference between the actual values and predicted values is as small as possible. The least Square Method is commonly used to find the best-fitting line.

For example, if we have data about hours studied and the marks obtained, we can use Simple Linear Regression to predict a student's marks based on the number of hours studied.

The general steps involves :-

1. Collect the data for the independent and dependent variables.
2. Divide the data into input (X) and Output (y).
3. Create a Linear Regression model.
4. Train the model using the available data.
5. Predict the output for new input values.
6. Visualize the regression line and data points
7. Evaluate the model using suitable performance measures.

* **Conclusion :-** Simple Linear Regression was successfully implemented using Python and the Scikit-learn library. The model established a linear relationship between hours studied and marks obtained. The trained model was also used to predict the marks for a new input value. The regression line helped to visualize the relationship between the independent and dependent var.

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Practical 3

Aim:- Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

* **Theory :-** Decision tree is a supervised machine learning algorithm used for both classification and regression problems. It represents the decision-making process in the form of a tree-structure. The tree consists of root node, internal decision nodes, branches, and leaf node.

For classification, a decision tree starts with a root node and divides the dataset into smaller groups based on the value of different attributes. The process continues until a suitable classification is obtained at the leaf node.

ID3 (Iterative Dichotomiser 3) is a decision tree algorithm developed by Ross Quinlan. It uses the concepts of Entropy and Information Gain to select the best attribute for splitting the dataset.

Entropy :-

Entropy measures the amount of uncertainty or impurity present in a dataset. It is calculated using :-

$$\text{Entropy}(S) = -\sum p_i \log_2(p_i)$$

where :- S = dataset

p_i = Probability of a particular class in dataset

* A higher entropy indicates more uncertainty.
* Entropy becomes 0 when all samples belong to the same class

* **Information Gain :-** Information Gain measures how much uncertainty is reduced after splitting the dataset using a particular attribute.

$$IG(S, A) = \text{Entropy}(S) - \sum \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

where,

* S = Original dataset
* A = Attribute used for splitting
* S_v = Subset created after the split
* $|S|$ = Total number of samples

The ID3 algorithm selects the attribute having the highest information Gain as the decision node.

* Working of ID3 Algorithm

1. Start with the complete training dataset
2. Calculate the entropy of the dataset
3. Calculate the IG for each available attribute
4. Select the attribute having the highest IG.
5. Make this attribute the decision node.
6. Divide the dataset according to the values

of the selected attribute.
⑨ Repeat the same process for the resulting subsets.
⑧ Continue until the data can be classified into the required classes.
⑨ The final leaf nodes represents the predicted classes.
⑩ Use the constructed trees to classify a new sample.

* Conclusion :- Thus the Decision tree based ID3 algorithm was successfully implemented using Python. The algorithm used entropy and inform gain to select the appropriate attributes for constructing the decision tree. The trained decision tree was then used to classify a new sample based on its attributes values. The result demonstrates how the ID3 algorithms can be used for classification problems using a decision tree structure.

Practical 4

* Aim:- Implementation of K-Means Clustering

* Theory:- K-Means Clustering is an unsupervised machine learning algorithm used to devid a dataset into a predefined number of groups called clusters. Unlike supervised learning, K-means does not require labelled output data.

The main data objective of K-means is to group similar data points together and keep dissimilar data points in different clusters.

The value of K represents the number of clusters that we want to create. For example, if K=3, the algorithm divides the given dataset into three clusters.

* Working of K-means Clustering:-
① Select the number of clusters K
② Randomly select K points as initial cluster centers, called centroids.
③ Calculate the distance between each data point and each centroid.
④ Assign each data point to the nearest centroid.
⑤ Recalculate the centroid of each cluster by taking the mean of all points belonging to

that cluster.
⑥ Repeat the assignment and centroid calculation process.
⑦ Continue until the centroids become stable or there is very little change in their positions.
⑧ The final groups obtained are the required clusters.
The commonly used distance measures in K-mean is Euclidean Distance:-

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

K-Means attempts to minimize the within-cluster sum of squares (WCSS), which represent the distance between the data points and their respective cluster centroids.

* Conclusion:- Thus, the K-means clustering algorithm was successfully implemented using Python and the Scikit learn library. The given dataset was divided into three clusters based on the similarity between data points. The algorithm calculated the clusters centroids and assigned each data point to its nearest cluster. A new data point was also classified into its appropriate cluster. The graphical representation helped to visualize the clusters and their centroids.

Practical 5

* Aim:- Implementation of KNN using sklearn.

* Theory:- The K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification and regression problems. It is a simple algorithm that predicts the class of a new data point based on the classes of its nearest data points.

KNN does not build a complex model during training. Instead, it stores the training data and uses it when a new data point needs to be classified.

The value of k presents the number of nearest neighbors considered for making the prediction. For example, if K=5, the algorithm finds the three nearest data points and determines the class of the new data point based on them.

* Working of KNN
① Select the value of K
② Calculate the distance between the new data point and all training data points.
③ Identify the K nearest data points.
④ For classification, count the classes of these nearest neighbors.
⑤ Assign the class that occurs most frequently among the neighbors.

⑥ For regression, the average value of the nearest neighbors is generally used as the prediction.
The most commonly used distance measure in KNN is Euclidean Distance :-

$$d = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

A suitable value of K is important. A very small value of K can make the model sensitive to noise, while a very large value can make the model less sensitive to local patterns.
Ex:- KNN can be used to classify flowers based on their characteristics such as:-
* Sepal length
* Sepal width
* Petal length
* Petal width.

* Conclusion :- Thus, the KNN algorithm was successfully implemented using Scikit-learn and used to classify Iris flower samples based on their features.

Practical 6

* Aim:- Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data set.

* Theory:- An Artificial Neural Network (ANN) is a machine learning model inspired by the working of biological neural networks. It consists of interconnected processing units called neurons that learns pattern from the input data and use the learned pattern to make predictions.
An ANN generally consists of three types of layers:-
① Input layer:- Receives the input features.
② Hidden layer - Performs calculations and learns patterns from the input data.
③ Output layer - Produces the final prediction or classification.
Each connection between neurons has a weight, each neuron may have a bias. During training, the weight and biases are adjusted so that the network produces better predictions.
* Backpropagation:-
Backpropagation is a commonly used algorithm for training Artificial Neural Networks. It works by calculating the error between the

actual output and the predicted output and then propagating this error backward through the network.
The main steps of Backpropagation are:-
① Initialize the weights and biases.
② Pass the input data through the network. This is called foward propagation.
③ Calculate the prediction output.
④ Compare the predicted output with the actual output and calculate the error.
⑤ Propagate the error backward through the network.
⑥ Calculate the gradients of the weights and biases.
⑦ Update the weights and biases using an optimization method such as gradient descent.
⑧ Repeat these steps for several iterations or epochs until the error is reduced.

* Conclusion:- Thus an Artificial Neural Network was successfully implemented using the Backpropagation algorithm and used to classify data set samples given

Practical 7

* Aim:- Performance analysis of Classification Algorithms on a specific dataset.

* Theory:- Classification is a supervised machine learning technique used to predict the class or category of a given data sample. In classification, a model is trained using labelled data and then used to predict the class of new or unseen data.
Different classification algorithms may produce different results on the same dataset. Therefore, performance analysis is useful for comparing their prediction accuracy and other evaluation measures.
For this practical, the Iris dataset can be used. It contains measurements of iris flowers and three different flower classes.
* Iris Setosa
* Iris Versicolor
* Iris Virginica
The features used are:-
* Sepal length
* Sepal width
* Petal length
* Petal width.
Classification Algorithms:-
① Logistic Regression - A statistical classification

algorithm that predicts the probability of a sample belonging to a particular class.
② K-Nearest Neighbors :- Classifies a new sample based on the classes of its nearest neighboring samples.
③ Decision tree :- Uses a tree-like structure of decisions to classify data.
④ Support Vector Machine (SVM) :- Finds a suitable decision boundary that separates different classes.
⑤ Random Forest :- Combines multiple decision trees to make a final prediction.

* Performance Evaluation :- The performance of classification algorithms can be evaluated using different metrics.
Accuracy measures the percentage of correctly ~~pre~~ classified samples.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Other commonly ~~pre~~ used measures are :-
* Precision :- Measures how many predicted positive samples are actually positive
* Recall :- Measures how many actual positive samples are correctly identified.
* F1-Score :- Represents the balance between precision and recall.

* Conclusion :- Thus, different classification algorithms were successfully implemented and their performance was analyzed using the Iris dataset based on accuracy, precision, recall, and F1-score.

Practical 8

* Aim :- Implement Naive Bayes theorem to classify the English text

* Theory :- The Naive Bayes is a supervised ~~learni~~ machine learning algorithm, based on Bayes' Theorem. It is commonly used for classification tasks, especially text classification.
It is called "Naive" because it assumes that the features of a sample are independent of each other given the class. Although this assumption is usually not completely true for natural language, the algorithm performs well for many text classification tasks.

Naive Bayes for Text classification :-
In text classification, the input is an English sentence or message. The algorithm first converts the text into numerical feature using a technique called as Count Vectorization.
For example :-
* "Win Free prize"
The words in this sentence are converted into numerical representations. The Naive Bayes classifier then learns the relationship between the words and their ~~co~~ corresponding classes.

For example, messages can be classified into :-
* Spam
* Not Spam.
When a new message is given, the classifier calculates the probability of each class and assigns the message to the class having the highest probability.

* Working Steps :-
① Collect a labelled English text dataset
② Divide the dataset into training and testing data
③ Convert the text into numerical features.
④ Train the Naive Bayes classifier.
⑤ Give a new English text to the trained model.
⑥ Calculate the probability of each class.
⑦ Assign the class with the highest probability.
⑧ Evaluate the performance of the classifier.

* Conclusion :- Thus, the Naive Bayes algorithm was successfully implemented to classify English text into Spam and Not Spam categories.